RAG-Explainable Predictive Maintenance Engineering Manual

This manual provides a comprehensive guide for integrating Explainable Artificial Intelligence (XAI) and Retrieval-Augmented Generation (RAG) to predict and explain industrial machine failures. It is derived from the AI4I 2020 Predictive Maintenance Dataset and research by Matzka (2020).

1. System Overview

The system integrates predictive modeling with explainability logic for human-interpretable outputs. It combines a Bagged Tree ensemble classifier with explainable decision trees and normalized feature deviations.

2. Dataset Specification

The dataset contains 10,000 data points, each representing machine operational conditions. Features include Product ID, Air Temperature, Process Temperature, Rotational Speed, Torque, Tool Wear, and a failure label.

Failure Mode	Abbrev.	Condition	Frequency
Tool Wear Failure	TWF	Tool wear > 200–240 min	120
Heat Dissipation Failure	HDF	(Proc T – Air T) < 8.6 K and RotSpeed < 1380 rpm	115
Power Failure	PWF	Power outside [3500, 9000] W	95
Overstrain Failure	OSF	Torque × Tool Wear > 11k–13k min·Nm	98
Random Failure	RNF	Random 0.1% probability	19

3. Explainability Framework

Two explanation mechanisms are used: (1) Decision Trees with ≤4 nodes for rule-based interpretability, and (2) Normalized Feature Deviation (NFD) for model-agnostic explanations. Both approaches enhance transparency in predictions for maintenance engineers.

4. Integration into RAG Pipeline

This manual should be embedded as structured knowledge for the RAG retriever. Each section can be chunked into ~200-token atomic contexts describing thresholds, rules, or interpretations. These enable human-readable, contextual predictions during maintenance queries.

5. Example Maintenance Report Template

MACHINE ID: H35754
PREDICTED STATUS: FAILURE (Prob: 0.92)
PRIMARY CAUSE: Power Failure
EVIDENCE:
- Torque: 72.8 Nm (High)
- Rotational Speed: 1324 rpm (Low)
INTERPRETATION: Power output below safe threshold (3500 W).
RECOMMENDATION: Inspect motor coupling and replace worn tool.

6. Future Extensions

- Integrate LIME or SHAP for localized interpretability.
- Apply Confidence Scores to quantify explanation trust.
- Deploy multilingual output (English/Japanese) for industrial use.